

Laxmi Shravani Mamidala

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GenAI Engineer | Data Scientist | LLM, RAG, MLOps & Applied AI Engineer

Professional Summary

AI Specialist with an M.S. in Artificial Intelligence and 5+ years of experience in AI, data science, and software engineering. Skilled in building production-ready LLM agents, RAG pipelines, APIs, automation frameworks, and developer tooling across the Software Development Lifecycle. Known for analytical thinking, writing skills, mentorship, and the ability to lead cross-functional AI and software engineering initiatives from concept to deployment.

Technical Skills

Languages and CS: Python, Java, SQL, R, JavaScript, HTML, CSS, Computer Science, Algorithms, Data Structures, Abstraction, Git, Unix, Command Line

AI, ML, and Data Science: Machine Learning, Deep Learning, NLP, Transformers, Predictive Modeling, Classification, Regression, Reinforcement Learning, DQN, Feature Engineering, Statistical Analysis, Data Generation, Model Evaluation, Benchmarking

Generative AI and LLMs: Large Language Models, LLMs, GPT, Gemini, LLaMA, Prompt Engineering, RAG, Embeddings, Vector Search, FAISS, LangChain, LangGraph, LangSmith, CrewAI, Tool Calling, Function Calling, Structured Outputs, Context Management

Cloud, APIs and MLOps: AWS, Azure, GCP, Cloud Architecture, Web Services, REST APIs, FastAPI, Flask, Docker, Kubernetes, MLflow, CI/CD, Model Deployment, Monitoring, Scalability, Reliability, Automated Testing, TDD, Test-driven Development, Observability

Explainability and Tools: SHAP, LIME, Grad-CAM, Metadata, Ontology, Data Modeling, ETL, Dash, Plotly, Jupyter, JIRA, Agile, Troubleshoot, Root Cause Analysis,

Professional Experience

University of North Texas, Denton, TX

AI Engineer

May 2025 – May 2026

- Led planning and execution of AI-native transformation initiatives, designing multi-agent applications with LangGraph and CrewAI to automate complex engineering workflows, accelerate developer enablement, and demonstrate the ability to lead reusable AI tooling adoption.
- Architected scalable AI web services and real-time inference APIs using FastAPI and cloud-native architecture; enforced reliability through TDD, automated testing, CI/CD, and full SDLC practices while developing LLM evaluation and benchmarking pipelines for retrieval quality and production scalability.
- Provided mentorship and senior technical enablement on GenAI workflows, prompt engineering, RAG implementation, and repository-level AI tooling; authored technical specifications, skill configurations, workflow documentation, and reference implementations for ontology-aware retrieval systems and reusable RAG components.

Data Scientist

August 2024 – May 2025

- Developed end-to-end data science and ML pipelines using Python, SQL, and TensorFlow, owning requirements through model deployment, monitoring, and SDLC execution; applied analytical thinking to optimize feature engineering and statistical workflows, improving accuracy, precision, and F1-score.
- Processed 10,000+ records for predictive modeling and text analytics; used SHAP/LIME to produce explainable insights, integrated ML models into scalable APIs and dashboards, and used communication skills, writing skills, and the ability to clarify technical trade-offs for leadership.

Infosys Limited, Hyderabad, India

Technology Analyst

June 2018 – July 2023

- Delivered enterprise AI and software engineering solutions for large-scale digital platforms, operating across requirements, architecture, build, test, deploy, and delivery in Agile/JIRA environments.
- Applied diverse programming skills to architect scalable, production-ready systems and contributed to principal-level technical initiatives and design reviews.
- Implemented code generation strategies and automation to enhance developer productivity and reduce repetitive manual coding tasks.Led the architecture and development of robust software systems, improving performance and maintainability by 30%.
- Implemented best practices across the Software Development Lifecycle, enhancing product reliability and performance
- Architected the IDSML platform to automate AI workflows, experimentation, and benchmarking, driving adoption of internal tooling and standard practices across engineering teams.
- Designed scalable microservices and ETL pipelines with reusable abstractions to improve operational excellence and metadata handling for production data processing.
- Automated Python-based enterprise workflows across 300+ network devices, improving troubleshooting speed and repeatability of command-line operations.
- Provided technical leadership as Scrum Master—conducting code reviews, hands-on pairing, and mentoring to raise engineering standards and secure data handling practices.

Projects

ProComp: Multi-Agent LLM Framework for Scholarly Evaluation

May 2025 – May 2026

- Built a multi-agent LLM evaluation system simulating peer review using RAG, role-based agents, and structured outputs to automate scholarly analysis across 190+ papers.
- Implemented explainable scoring and ranking algorithms to ensure reliability and consistency in automated AI-native workflows.

Agentic AI for Enterprise Conversation Analysis and Workflow Automation

May 2025

- Developed a CrewAI-based pipeline for enterprise conversation intelligence, integrating transcription, metadata extraction, and REST APIs.
- Designed automated agent workflows to bridge the gap between raw data and actionable enterprise insights.

Real-Time Object Detection and Deep Learning Inference System

Aug 2025

- Built a YOLO-based object detection pipeline with Flask/FastAPI deployment and real-time inference performance tuning.Optimized deep learning models using Grad-CAM for explainability and architectural guidance in computer vision tasks.

Deep Reinforcement Learning for Decision Optimization using DQN

Jan 2025

- Implemented a DQN model using neural networks and reward optimization, achieving a 35% improvement over baseline strategies.Analyzed convergence and exploration trade-offs to optimize decision-making in ambiguous environments.

Education

Master of Science in Artificial Intelligence,

University of North Texas

August 2024 to May 2026

GPA: 4 /4 | Distinguished MS Scholar Award

Bachelor of Technology in Computer Science Engineering,

Sri Indu College of Engineering and Technology

August 2014 to May 2018

GPA: 4/4 | 2x Academic Excellence Award

Certifications and Awards:

- Distinguished MS Scholar Award
- Udacity AWS AI Practitioner Challenge
- Infosys Certified Python Associate
- 2x Academic Excellence Award